CEBU INSTITUTE OF TECHNOLOGY - UNIVERSITY
College of Computer Studies, IT411 Capstone & Research 2

Final User & Stakeholder Validation Plan and Framework

Deliverable 04: research questions, cohorts, evaluation models, and statistical methods

Course & Section:	IT411 (Section G7) — Capstone & Research 2 | Semester 1, AY 2026–2027
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Deliverable Reference:	Pre-Adviser Presentation Requirement D (`Final_Validation_Plan_and_Framework`) & Week 12 Deliverable 1 (`01_Validation_Plan_and_Framework`, Due October 31, 2026)
System Version Under Test:	`playIT v1.2.0-rc2` (`refactor/hear-say-it` post-Card 32 production build)

1. Validation Purpose & Objectives

Unlike the initial **Weeks 1–2 MVP Validation** (which diagnosed early prototype bottlenecks on a baseline build), this **Final User/Stakeholder Validation** evaluates the **completed, refactored, and deployed playIT Android system** (`v1.2.0-rc2`) to determine with empirical evidence whether the system:

1. Effectively addresses the identified early-reading literacy problem for Grade 1 learners (ages 6–7);
2. Meets the functional, pedagogical, and ergonomic needs of its intended end-users (Grade 1 learners) and adult stakeholders (DepEd Grade 1 reading teachers and parents/caregivers); and
3. Achieves every measurable target specified in the group's approved SMART goals (`SMART-01` through `SMART-07` and operational sub-objectives `HI-1`, `HI-2`, `SI-1`–`SI-4`, `O3`–`O6`).

1.1 Research Questions (`RQ1`–`RQ5`)

Research Question ID	Research Question Statement	Linked SMART Goals
`RQ1` **(Pedagogical & Acoustic Integrity)**	To what extent do playIT's 26 phoneme audio models, 7 articulatory mouth-shape groups, and 33 Marungko-aligned decodable CVC/VVC words satisfy DepEd Grade 1 phonics standards and Gate 3 acoustic purity criteria?	`SMART-01`, `SMART-02`, `HI-1`, `HI-2`
`RQ2` **(Speech Recognition & Scaffolding Effectiveness)**	Does the refactored offline Vosk `SpeechValidator` achieve ≥ 80% agreement with blind human teacher ratings and ≤ 15% false rejections on Grade 1 child speech within ≤ 500 ms while maintaining zero punitive heart loss?	`SMART-03`, `SI-1`, `SI-2`, `SI-3`, `SI-4`
`RQ3` **(Child Usability, Discrimination & Blending Mastery)**	Can Grade 1 learners (ages 6–7) independently navigate playIT's onboarding, Bohol adventure map, `Hear It`, `Say It`, `Find It`, and `Blend It` stages (≥ 85% zero-prompt transitions) and achieve ≥ 2star mastery with positive affect?	`SMART-04`, `SMART-05`, `O3`
`RQ4` **(Hardware Performance, Offline Persistence & Privacy)**	Does the deployed APK meet all real-time latency (≤ 500 ms Say It, ≤ 300 ms Find It), frame stability (≤ 10% jank at 60 FPS), 20-cycle force-close SQLite persistence, and Airplane-Mode sovereignty requirements on budget hardware (`SM-A217F`)?	`SMART-06`, `O4`, `O5`
`RQ5` **(Stakeholder Usability, Diagnostic Utility & Acceptance)**	Do Grade 1 reading teachers and parents/caregivers rate playIT's overall usability at SUS ≥ 75.0 and perceived usefulness/ease-of-use at TAM ≥ 4.00/5.00, and can they accurately interpret the Parent Zone and exported A4 PDF diagnostic report?	`SMART-07`, `O6`

2. Target Respondents & Sampling Justification

To prevent single-source bias and ensure that **child behavioral outcomes** are evaluated separately from **adult diagnostic and pedagogical outcomes**, the validation employs a purposive, multi-stakeholder sample across four cohorts (N_(total) = 31–38 participants):

Cohort ID	Respondent Group	Target Sample Size	Inclusion Criteria & Relevance to Intended Users/Stakeholders	Assigned Validation Activities & Instruments
Cohort 1 (`C-01..C-20`)	**Grade 1 Learners (Primary End-Users)**	**N = 16–20 children** (generating ≥ 150–240 logged `Say It` & `Find It` trials)	Enrolled Grade 1 learners aged **6–7 years old** (early/emergent readers). Directly tests whether pre-readers can operate the 64dp interface, articulate phonemes, discriminate letters, and blend CVC words without adult help.	Guided hands-on play session (15–20 min): Avatar onboarding → Group 1 letters (`M, S, A`) → `Blend It` chest; `Instrument B` **(Part B1)** Silent-Observer Sheet; `Instrument C` Room DB Telemetry; `Instrument E` **(Part E1)** Child Picture Choice & Smileyometer.
Cohort 2 (`T-01..T-05`)	**DepEd Grade 1 Reading / MTB-MLE & English Teachers (Domain Experts)**	**N = 4–5 teachers**	Licensed elementary educators with ≥ 2 years experience teaching Grade 1 reading/phonics (Marungko / MATATAG curriculum). Qualified to judge phoneme purity (schwa intrusion) and rate child pronunciations blindly.	1. `Instrument A` Blind 26-Phoneme Audit (Gate 3) & 33-Word CVC Audit; 2. `Instrument B` **(Part B2)** Blind Dual-Rater scoring of child `Say It` utterances; 3. `Instrument E` **(Part E2)** Teacher SUS + TAM + Diagnostic Report Evaluation.
Cohort 3 (`P-01..P-10`)	**Parents / Primary Caregivers (Adult Stakeholders)**	**N = 8–10 parents/guardians**	Parents or guardians of Grade 1 children who supervise home reading practice on shared smartphones. Evaluates profile management, the arithmetic gate, progress interpretation, and offline PDF sharing.	Hands-on Parent Zone task walkthrough (unlocking `ArithmeticGuardDialog`, renaming profile, identifying `At-Risk` letters, exporting & opening A4 PDF) + `Instrument E` **(Part E2)** Parent SUS + TAM Survey.
Cohort 4 (`QA-01..QA-03`)	**Technical / QA Evaluators**	**N = 3 evaluators**	Capstone technical evaluators executing objective hardware benchmarks and accessibility audits on the physical **Samsung Galaxy A21s** (`SM-A217F`) test device.	`Instrument D` (`D-01` to `D-12`): ADB latency/frame/memory benchmarks, 20× force-close persistence in Airplane Mode, 4-viewport bounds check.

3. Proposed Validation Model & Theoretical Framework

Our validation architecture integrates four complementary, internationally recognized frameworks so that every SMART goal is evaluated through the right methodological lens:

```
flowchart LR
subgraph Framework["Triangulated Final Validation Framework"]
direction TB
F1["1. Objective-Driven Behavioral & Acoustic Audit\n(Gate 3 Phoneme Audit + Blind Teacher-ASR Agreement)"]
F2["2. ISO 9241-11:2018 Usability Standard\n(Effectiveness, Efficiency, Satisfaction)"]
F3["3. Pediatric UPA Diagnostic Model\n(Usability, Pedagogy, Accessibility / Read & MacFarlane)"]
F4["4. Technology Acceptance Model (TAM) & Brooke SUS\n(Perceived Usefulness, Ease of Use, System Usability Scale)"]
end
F1 --> M1["Measures: SMART-01, SMART-02, SMART-03 (HI-1, SI-1..SI-3)"]
F2 --> M2["Measures: SMART-04, SMART-06 (Task Completion, P90 Latency, Persistence)"]
F3 --> M3["Measures: SMART-05, O3 (Zero-Prompt Child Nav, 64dp Ergonomics, Affect)"]
F4 --> M4["Measures: SMART-07, O6 (Parent/Teacher SUS >= 75, TAM >= 4.00)"]
```

1. **Objective-Driven Acoustic & Psychometric Agreement Model (Signal Detection / Cohen's kappa):**
   - Evaluates `SMART-02` (`HI-1`) and `SMART-03` (`SI-1`–`SI-3`). Rather than asking adults whether speech recognition "seems accurate" on a questionnaire, two teachers blindly score each child utterance (T in {0, 1}), and their consensus is cross-tabulated against the offline Vosk `SpeechValidator` decision (A in {0, 1}) to compute exact **Teacher-ASR Agreement (P_(agree))**, **False-Reject Rate (P_(FR))**, and **Cohen's/Fleiss' kappa**.
2. **ISO 9241-11:2018 Ergonomics of Human-System Interaction:**
   - **Effectiveness:** Task completion rates across child stages (`Hear It`, `Say It`, `Find It`, `Blend It`) and adult tasks (Parent Gate, PDF Export).
   - **Efficiency:** Objective system response times (`Say It` P90 latency ≤ 500 ms, `Find It` P90 latency ≤ 300 ms, lesson completion time in seconds) captured via `Instrument C` and `Instrument D`.
   - **Satisfaction:** Child Smileyometer & voluntary replay rate (`Instrument E` Part E1) and adult **System Usability Scale (SUS)** (`Instrument E` Part E2).

3. **Pediatric UPA (Usability, Pedagogy, Accessibility) Framework & Read & MacFarlane (2006) Child-Computer Interaction Protocol:**

- Standard Likert surveys are developmentally invalid for 6-to-7-year-old pre-readers. Following Read & MacFarlane (2006), child evaluation relies on **Silent-Observer Behavioral Coding (**`Instrument B`**)**, a **3-Point Visual Smileyometer**, a **Picture-Card Hardest-Stage Identification**, and a **Behavioral Replay Prompt** (*"Would you like to play one more letter or stop now?"*).

4. **Technology Acceptance Model (TAM — Davis, 1989) & System Usability Scale (SUS — Brooke, 1996):**

- Administered to Teachers and Parents (`Instrument E` Part E2) to quantify **Perceived Usefulness (PU)**, **Perceived Ease of Use (PEOU)**, **Behavioral Intention (BI)**, and standardized **SUS (0–100)**.

4. Evaluation Procedures & Step-by-Step Activities

Phase 1: Pre-Field Gate 3 Teacher Acoustic & Content Audit (`Instrument A`)

1. Prior to child sessions, **3–5 DepEd Grade 1 teachers** independently complete `Instrument A`.

2. In **Part A1 (Blind Phoneme Audit)**, all 26 `audio/phonemes/{letter}.mp3` clips are played in randomized order without showing the letter label. Teachers write the letter sound they hear and rate clarity (1 = Unacceptable / Has schwa, 2 = Acceptable, 3 = Model quality). All 26 clips must pass before child testing.

3. In **Part A2 & A3**, teachers audit the 33 decodable CVC/VVC words and rate curriculum alignment (`PA1`–`PA5`).

Phase 2: Hardware, Offline & Accessibility Verification (`Instrument D`)

1. On the physical **Samsung Galaxy A21s (**`SM-A217F`**, Android 12, 3 GB RAM)** with **Airplane Mode ON** and Wi-Fi/cellular disabled:

- Execute cold-start timing (`adb shell am start -W`), 60 FPS frame rendering (`dumpsys gfxinfo`), memory profiling (`dumpsys meminfo`), and a 5,000-event `monkey` stress run.
- Measure 30 trials of `Say It` end-of-speech to feedback latency (≤ 500 ms) and 30 trials of `Find It` tap-to-feedback latency (≤ 300 ms).
- Execute the **20-Cycle Force-Close Persistence Test** (`D-05`) and record all 12 engineering checks on `Instrument D`.

Phase 3: Child End-User Sessions — Silent-Observer Protocol (`Instrument B` & `Instrument E` Part E1)

1. **Ethics & Consent:** Written parental consent and verbal child assent are secured prior to every session. Each learner is assigned an anonymous code (`C-01`..`C-20`); zero names, photos of faces, or raw voice files leave the local session.

2. **Silent-Observer Rule:** The facilitator hands the device (in Airplane Mode) to the child on `SplashScreen` and sits silently beside them. The observer **only** intervenes if the child explicitly asks for help or makes no progress for ≥ 15 seconds (after the app's built-in 10s idle re-prompt has fired), tallying every adult prompt on `Instrument B` (Part B1).

3. **Standard Child Task Scenario (15–20 min):**

- **Task C1 (Onboarding):** Tap `Start` on Splash → tap `+ New Player` → pick an animal friend avatar → tap `Let's Play` (0 typing required).
- **Task C2 (Map & Letter Lesson Loop):** Tap active Node `M` on the Bohol Map → complete `Hear It` (3 audio steps) → complete `Say It` (spoken echo + recall check) → complete `Find It` (3 picture-grid rounds) → view `LetterCompleteScreen` stars.
- **Task C3 (Subsequent Letters & Blend It):** Complete Letters `S` and `A` (or transition to an unlocked Group 1 `Blend It` treasure chest) and blend 3 decodable CVC words (`AM`, `SAM`, `MAT`).

4. **Live/Recorded Teacher Rating of** `Say It` **Trials (**`Instrument B` **Part B2):** For each `Say It` attempt, two teachers independently mark whether the child produced the valid target sound/word (T = 1) or an invalid foil/silence (T = 0), logged alongside the app's `SpeechValidator` outcome (A in {0, 1}).

5. **Post-Session Child Exit Check (**`Instrument E` **Part E1):** The child points to the **Smileyometer** face, points to which of the 4 game picture cards felt hardest, and responds to the voluntary replay prompt (*"You're done! Would you like to play one more letter, or stop now?"*).

Phase 4: Parent & Teacher Diagnostic Walkthrough (`Instrument E` Part E2)

1. Parents (`P-01`..`P-10`) and Teachers (`T-01`..`T-05`) perform four concrete adult tasks on the device populated with session data:
   - **Task A1:** Open `Parent Zone` and solve the randomized `ArithmeticGuardDialog` challenge.
   - **Task A2:** Rename a learner's profile in `LearnerHeroCard` and switch between learner profiles.
   - **Task A3:** Inspect `MasteredSoundsShelf` and `PracticeFocusSection` to correctly name which letters are flagged as **Mastered (`Green`)**, **Developing (`Yellow`)**, or **At-Risk (`Red/Amber`)**.
   - **Task A4:** Tap **PDF**, preview the generated report on `ReportPreviewScreen`, and open the offline A4 PDF document.
2. Immediately after completing Tasks A1–A4, adult participants complete **Instrument E (Part E2)** (10-item Brooke SUS + 10-item TAM & Diagnostic Utility scale + 3 open-ended diagnostic prompts).

5. Statistical Analysis Methods & Decision Rules

Metric Category	Statistical Estimator / Calculation Formula	Acceptance / Success Criterion
Binomial Proportions *(Task completion, Zero-prompt navigation, Teacher-ASR agreement, False-reject rate)*	Point estimate hatp = k/n with **Wilson 95% Confidence Interval**: $$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96$$	hatp_(nav) ≥ 85.0%; hatp_(agree) ≥ 80.0%; hatp_(FR) ≤ 15.0%.
Inter-Rater Reliability *(Dual-teacher scoring of child* `Say It` *attempts)*	**Cohen's Kappa (kappa)** (2 raters) or **Fleiss' Kappa** (≥ 3 raters): $$\kappa = \frac{P_o - P_e}{1 - P_e}$$	kappa ≥ 0.70 (substantial inter-rater agreement).
Response Time & Latency Distributions *(*`Say It` *&* `Find It` *latency, Cold start)*	Median (**P50**), **90th Percentile (P90)**, Mean, Standard Deviation (SD), and Min–Max range in milliseconds.	**Say It P90 ≤ 500 ms**; **Find It P90 ≤ 300 ms**; **Cold Start ≤ 2500 ms**.
System Usability Scale (SUS) *(Brooke, 1996 — 10 items)*	Odd items: $x_i - 1$; Even items: $5 - x_i$: $$\text{SUS} = 2.5 \times \left(\sum_{\text{odd}} (x_i - 1) + \sum_{\text{even}} (5 - x_i)\right) \in [0, 100]$$	**Mean SUS ≥ 75.0** (Bangor et al. "Good" to "Excellent" adjective rating).
TAM & Pedagogical Likert Constructs *(5-point Likert items)*	Construct Mean barx = 1/N Σ x_i and Standard Deviation s: 4.21–5.00: Strongly Agree / Highly Effective 3.41–4.20: Agree / Effective < 3.41: Needs Improvement	**Mean ≥ 4.00 / 5.00** across Perceived Usefulness (PU), Perceived Ease of Use (PEOU), and Pedagogical Alignment (PA).